

1. Objective

To design and build an automation layer in Zoho Creator for the existing Event Registration application, demonstrating the Workflow options Zoho Creator provides (Form Workflows, Schedules, Approvals, Payments, Blueprints, Batch Workflows, Report Workflows and Functions) so that registration runs with minimal manual intervention from submission to check-in.

2. Problem Statement

The Event_Registration_Form captures data well, but downstream actions depend on people: sending confirmations, chasing pending payments, approving VIP passes, verifying documents, regenerating QR codes and recalculating fees. This does not scale beyond a few hundred registrations and causes delay and inconsistency. The automation suite routes routine actions automatically, sends exceptions for approval, and moves each registration through a controlled, auditable process.

3. Workflow Mapping

Workflow Option	Workflow Name	Purpose / Trigger
Form Workflow	On_Registration_Submit_Workflow	Instant confirmation email on form submission
Schedule	Daily_Payment_Reminder_Schedule	Daily reminder to participants with pending payment
Approval	VIP_Sponsorship_Pass_Approval	Coordinator approval for VIP/Sponsor passes
Payments	Registration_Fee_Payment_Workflow	Payment link for Total_Amount_Payable; updates Payment_Status
Blueprint	Registration_Lifecycle_Blueprint	Controlled stages from Submitted to Checked-In
Batch Workflow	Bulk_QRCode_Regeneration_Batch	Regenerate QR codes in batches of 200
Report Workflow	OnEdit_Recalculate_Fee_Workflow	Recalculate fee when fee/discount edited in report
Function	Generate_Certificate_Function	Reusable certificate email function with input validation

4. Form Workflow - On_Registration_Submit_Workflow

Fires on successful submission of the Register_For_Events form (Created > Successful form submission) and executes without condition.

- Action 1: *Send mail to \${Email}* - acknowledgement to the participant.
- Action 2: a Deluge script sends a "Participation Confirmation" email using input.Name and input.Event_Name.
- Governance: failures in email must not block the registration.

```
sendmail
[
  from :zoho.adminuserid
  to :input.Email
  subject : "Participation Confirmation for " + input.Event_Name + " Event"
  message : "Dear " + input.Name + ", ..."
]
```

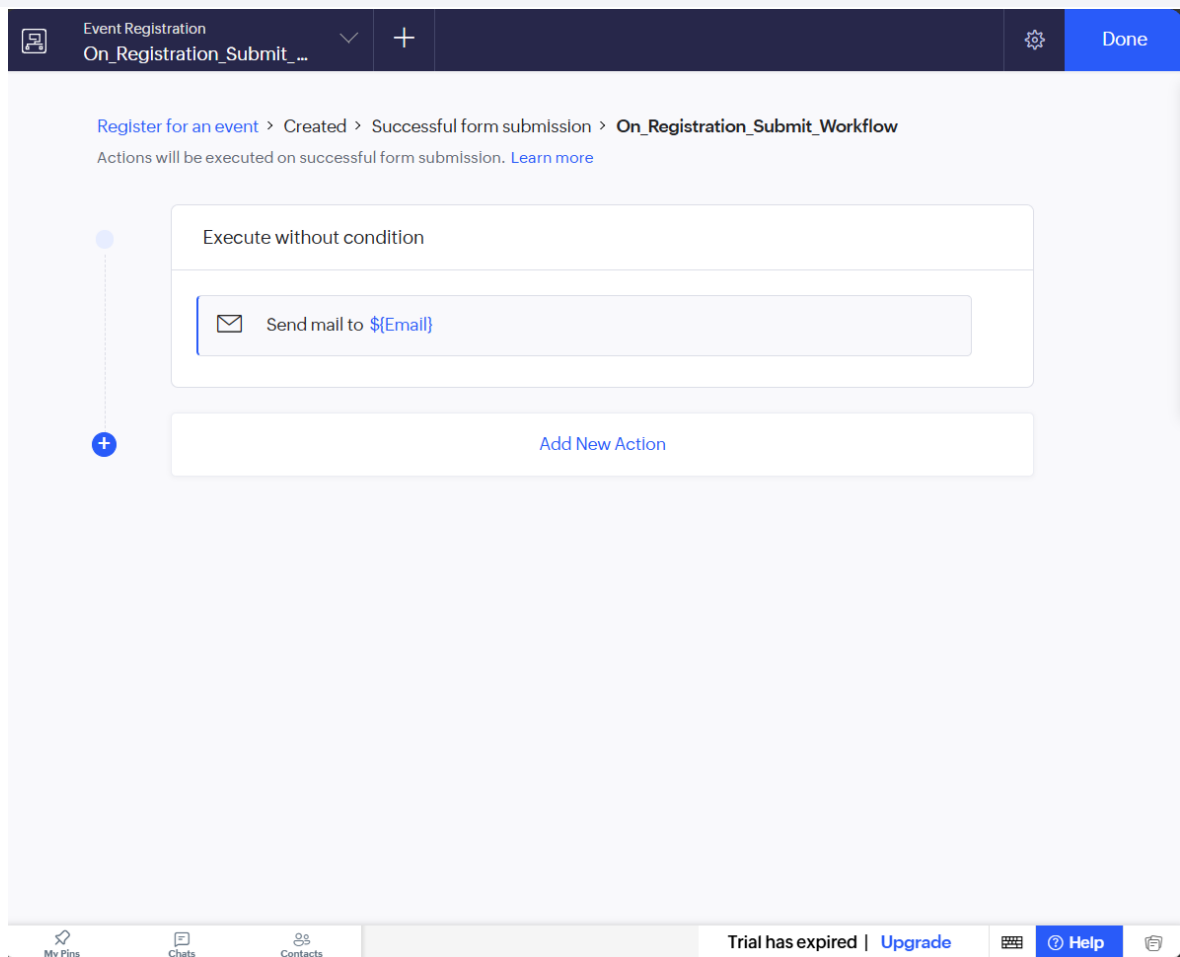


Figure 1: On_Registration_Submit_Workflow with Send mail action

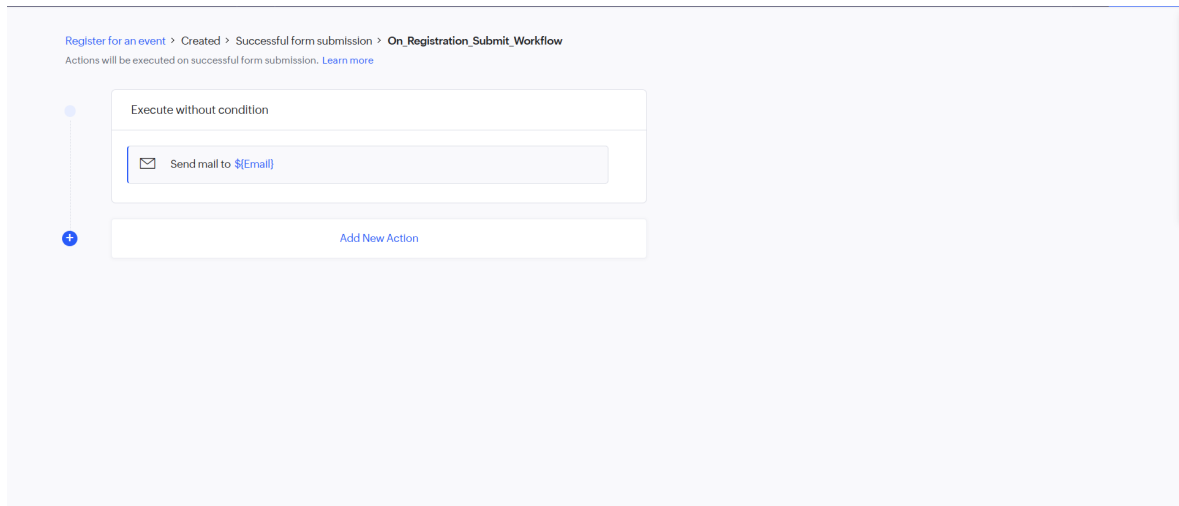


Figure 2: Workflow configuration view (full page)

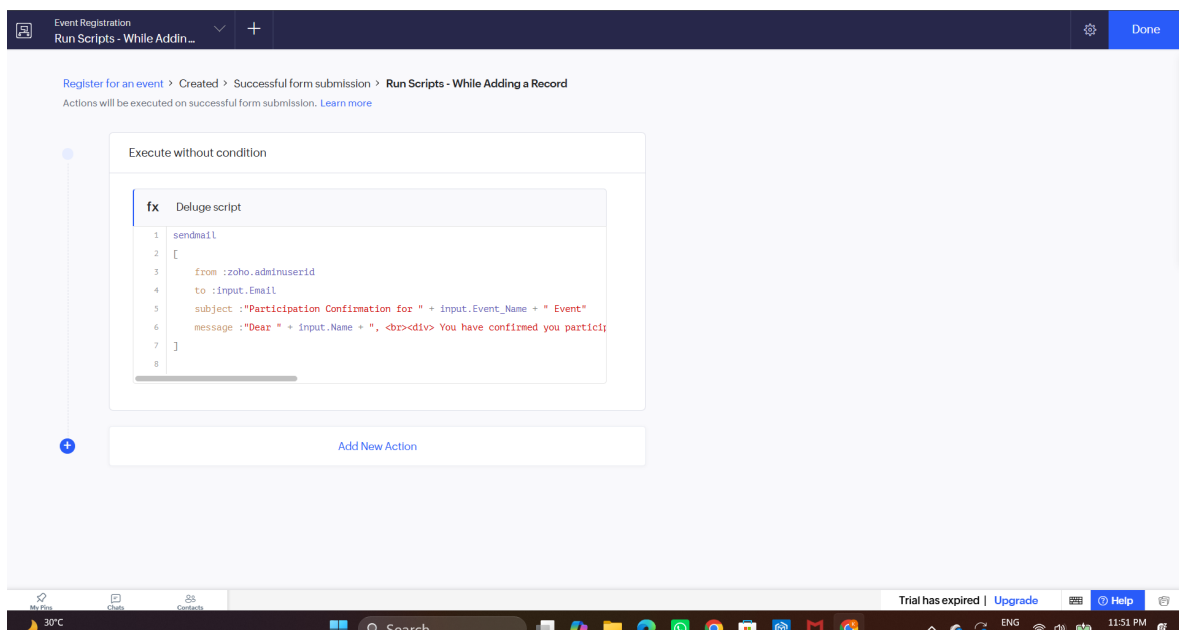


Figure 3: Deluge script action sending the confirmation email

5. Schedule - Daily_Payment_Reminder_Schedule

A custom schedule that runs once a day and loops over all registrations to email a payment reminder.

- Loops with *for each rec in Register_For_Events* and emails rec.Email.
- Subject: "Payment Reminder: Event Registration".
- Recommended refinement: filter on Payment_Status == "Pending", cap at 3 reminders per participant and stop after the closing date.
- The schedule is currently shown as **Disabled** in the Schedules list (created 09-Oct-2026, next run 10-Oct-2026 23:30:36).

```
for each rec in Register_For_Events
{
  sendmail
  [
    from :zoho.adminuserid
    to :rec.Email
    subject : "Payment Reminder: Event Registration"
    message : "Dear " + rec.Name + ",\n\nPlease complete your pending payment..."
  ]
}
```

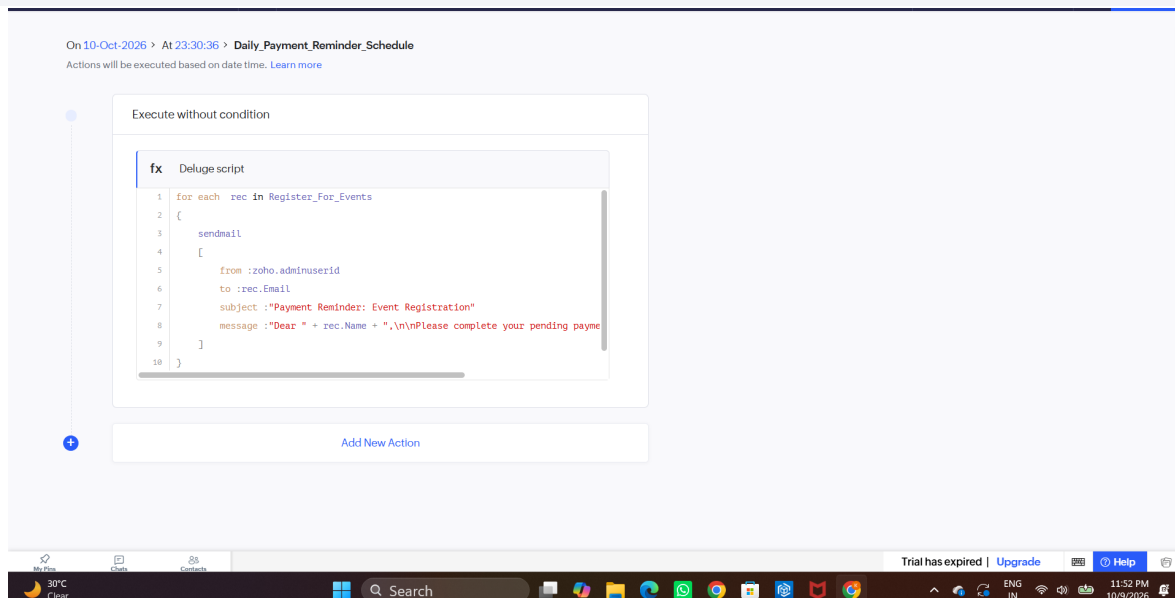


Figure 4: Schedule configuration with Deluge script

Event RegistrationWorkflow

DesignWorkflowSettings

Upgrade

Access this application

Form workflowsSchedulesApprovalsPaymentsBlueprintsBatch workflowsFunctions

Search for Workflows, Forms ...New Workflow

Workflow Name	Details	When	Status	Created On
Custom Schedules				
Daily_Payment_Reminder_Schedule	1 Action	10-Oct-2026 : 23:30:36	Disabled	09-Oct-2026

Trial has expired | Upgrade

Help

Figure 5: Schedules list showing Daily_Payment_Reminder_Schedule

6. Approval - VIP_Sponsorship_Pass_Approval

Routes VIP/Sponsor registrations to the Event Coordinator before the seat is confirmed. Regular registrations bypass it.

- Approver: Level 1 (single approver role, with a fallback approver recommended).
- On Approve: send mail to \${approver.submitter} and run a Deluge script.
- On Reject: send mail to \${approver.submitter} and run a Deluge script.
- Status is "Pending Approval" until the coordinator acts. The approval is Enabled (4 actions).

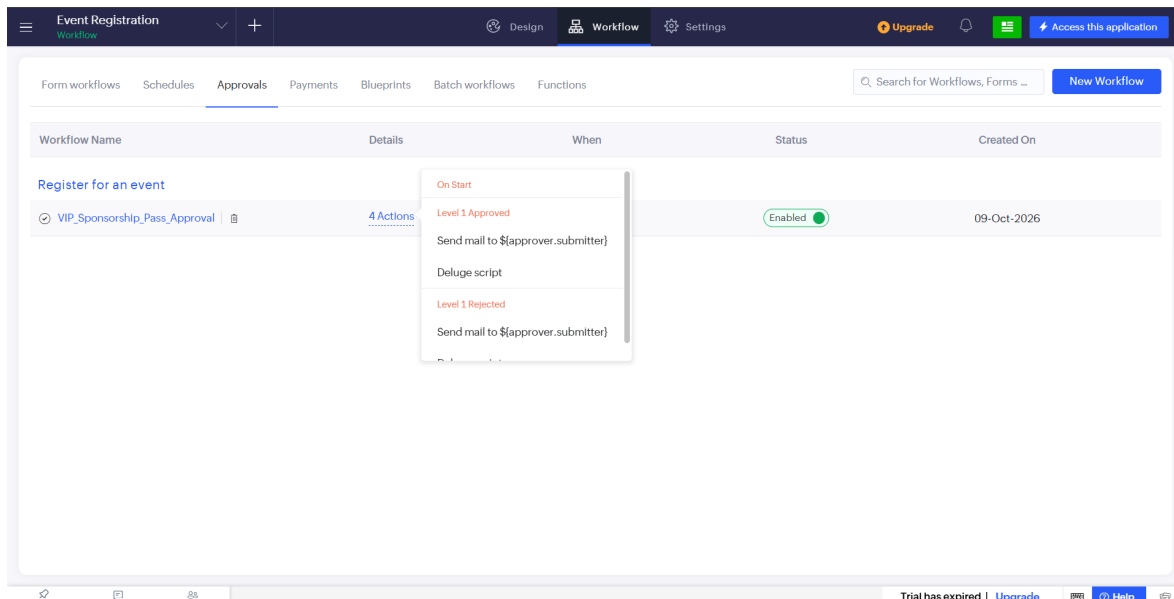


Figure 6: Approvals list with the 4 configured actions

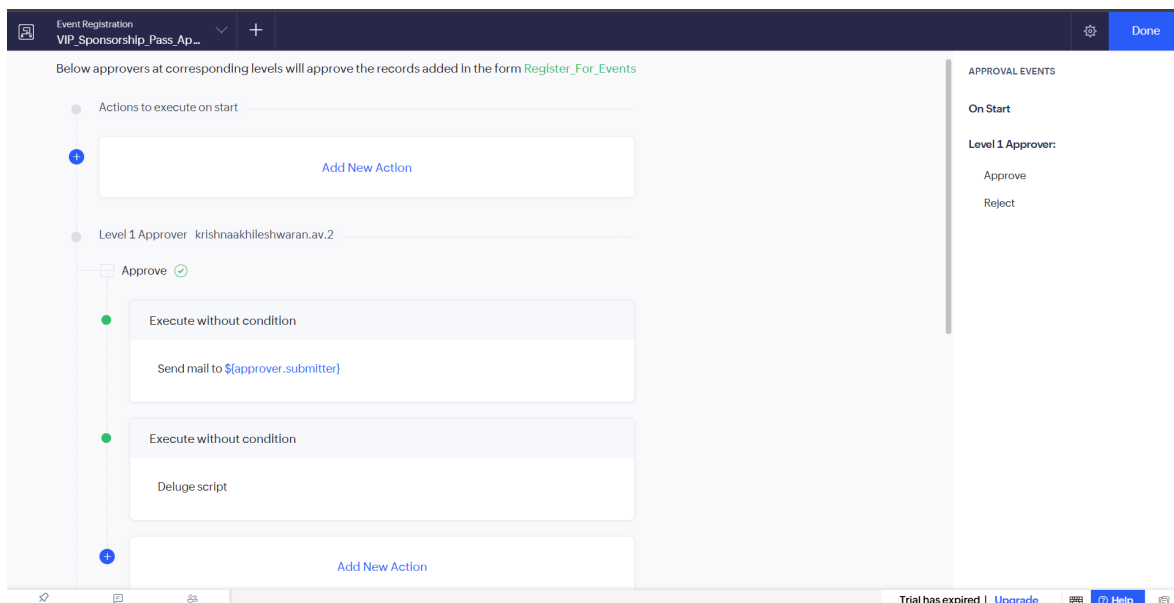


Figure 7: Approval configuration (Level 1 approve actions)

7. Blueprint - Registration_Lifecycle_Blueprint

Defines the controlled lifecycle of each registration as stages and transitions.

- Flow: Start > Submitted > Verify Payment > Payment Verification > payment > Document Verification > Confirm Seat > Confirmed > success > Checked-In.
- Dark blocks are transitions; light blocks are stages.
- Each transition should have role-based permissions and a mandatory comment for an audit trail.

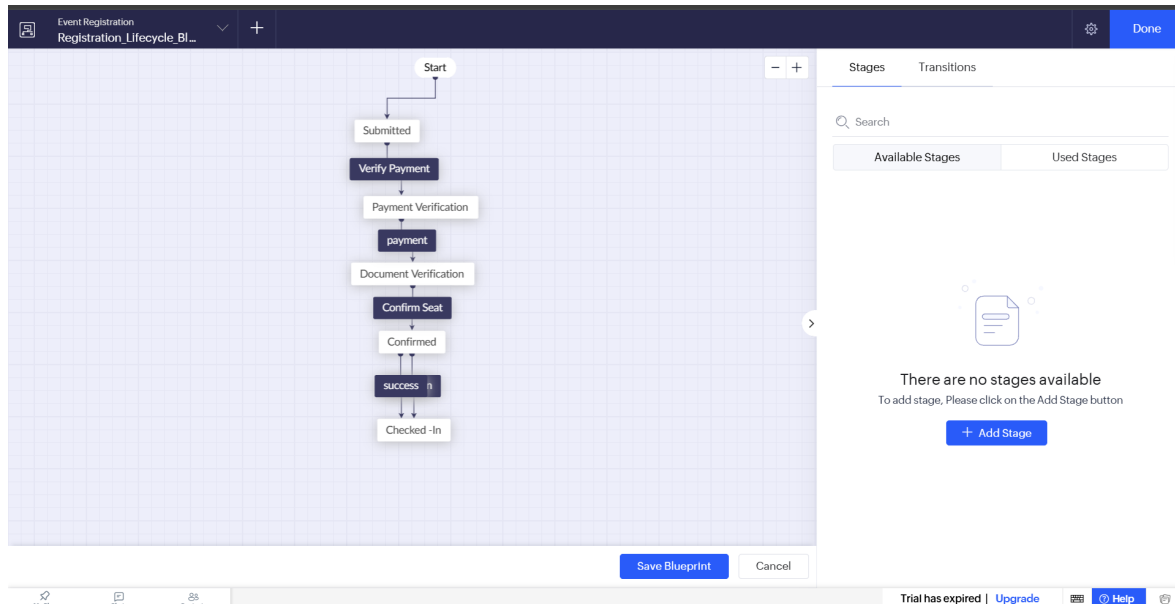


Figure 8: Blueprint stage diagram

8. Function - Generate_Certificate_Function

A reusable Deluge function that validates the record and emails a participation certificate.

- Return type: string; parameter: record ID (int).
- Input validation (governance rule): returns "Failed: Missing Name or Email" if Name or Email is empty - this was demonstrated in the Execute dialog (Figure 10).
- Sends "Your Event Participation Certificate" to rec.Email, otherwise returns "Success".
- Can be invoked from a Send Certificate button, a schedule, a batch, or an approval step.

```
string Generate_Certificate_Function(int recordID)
{
    rec = Register_For_Events[ID == recordID];
    if(rec.Name == null || rec.Email == null)
    {
        return "Failed: Missing Name or Email";
    }
    sendmail [ from:zoho.adminuserid to:rec.Email
              subject:"Your Event Participation Certificate"
              message:"Dear " + rec.Name + ",..." ]
    return "Success";
}
```

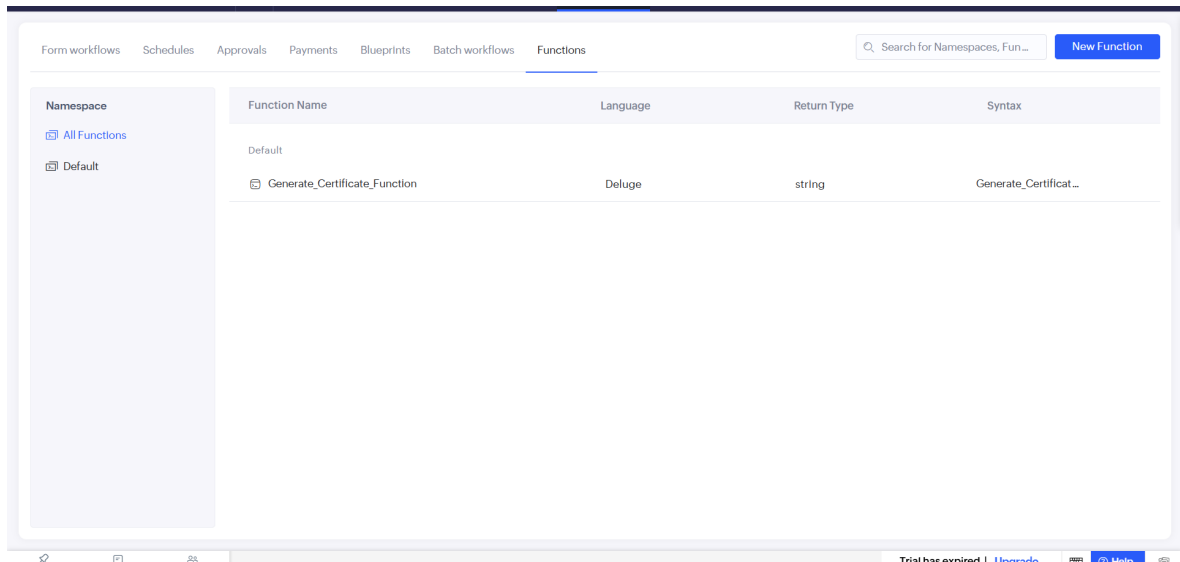


Figure 9: Functions tab listing Generate_Certificate_Function

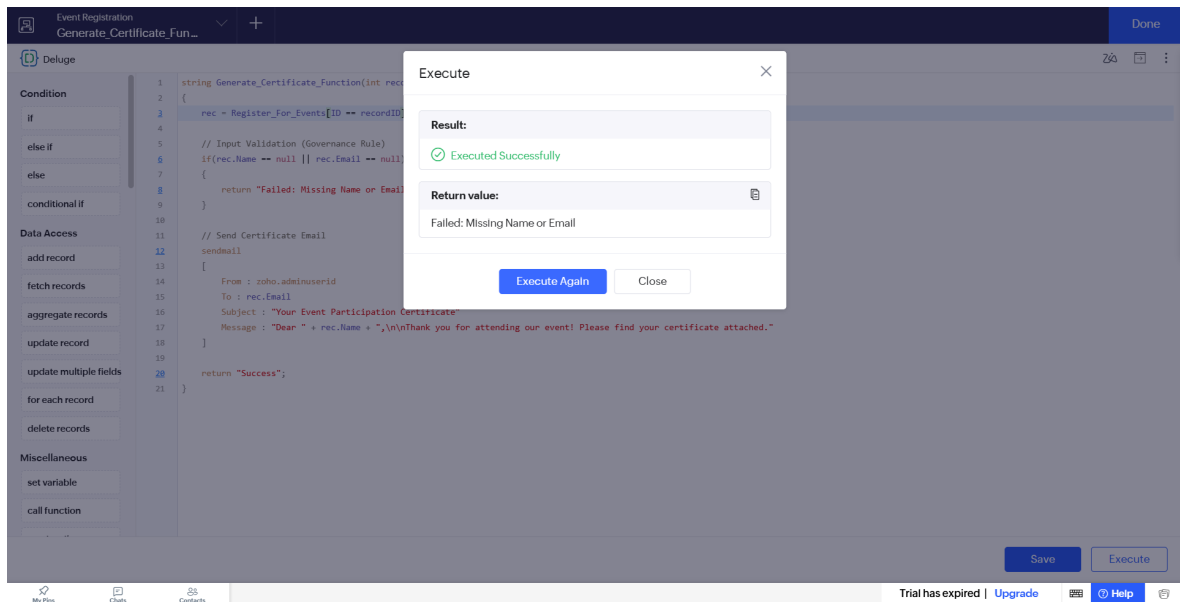


Figure 10: Function script and execution result (validation path)

9. Governance Rules Summary

- Form Workflow: fail gracefully on invalid email.
- Schedule: maximum 3 reminders per participant; stops after closing date.
- Approval: single approver role plus fallback approver.
- Payments: retry gateway status callback up to 3 times before flagging failure.
- Blueprint: mandatory comment on every transition.
- Batch: configurable batch size (default 200).
- Report Workflow: fire only when Registration Fee or Discount changes.
- Function: input validation for missing Name/Email.

10. End-to-End Summary

A participant submits the form and receives an immediate acknowledgement; payment is collected and Payment_Status updated; VIP/Sponsor records go to the coordinator for approval; staff move records through blueprint stages with logged comments; daily reminders chase pending payments; fee edits in the report are recalculated; QR codes are regenerated in batches before the event; and certificates are issued after the event through the reusable function.

11. Learning Outcomes

- Choosing the right workflow type for each use case.
- Event-driven (form, report) versus time-driven (schedule) automation.
- Combining Blueprints (process control) with Approvals (human checkpoint).
- Using batch workflows to avoid timeouts on large data volumes.
- Writing logic once as a Function and reusing it.

12. Conclusion

The Event Registration application now automates confirmation, reminders, approvals, lifecycle control and certificate delivery in Zoho Creator, reducing manual effort and giving the committee a consistent, auditable registration process.